# Sharp Stability Radii for Thresholded Packet Branches
## Clipped Singular-Value Selection and Projected-Cross Perturbations

Prime Dynamics Theory Program

July 2026

**Abstract**

Adaptive packet updates choose a width by comparing projected-cross singular values with a relative threshold. This branch is discontinuous at contact, so an interval packet chain requires an explicit stability radius. Let $s_1 \geq \cdots \geq s_M \geq 0$, choose

$$w = \max\{m, \min(M, \#\{j : s_j/s_1 \geq \tau\})\},$$

and perturb the cross operator by at most $\varepsilon$. We prove that the clipped width is unchanged whenever

$$\varepsilon < \varepsilon_* = \min\left\{\frac{s_w - \tau s_1}{1+\tau}\ (w > m),\quad \frac{\tau s_1 - s_{w+1}}{1+\tau}\ (w < M)\right\}.$$

The omitted terms at the clipping endpoints are genuinely irrelevant. Each constant is sharp by a diagonal singular-value perturbation, and the radius vanishes exactly at threshold contact.

For packet projectors we additionally prove

$$\|(I-P)AP - (I-Q)BQ\| \leq \|A-B\| + 2\|B\|\|P-Q\|.$$

This provides the interface from snapshot/projector balls to the singular branch radius. Auditing all three RH-96 thresholds gives 360/360 strict branches and exact agreement with the archived widths. Every branch also dominates a conservative local fp64 proxy $128un\|K\|$; the smallest ratio is above 2388. Nevertheless the primary minimum relative branch radius is only $4.3449 \times 10^{-9}$. Thus local numerical branching is robust, but a full source-to-update interval enclosure is not yet proved.

# 1 Why width selection needs its own theorem

RH-142 fixes the rank-four source projector at all ten frozen anchors. The recursive packet algorithm has another discontinuity: it selects two, three, or four enrichment directions from the singular values of a projected cross. A small perturbation can change this integer branch even when the underlying packet subspace varies continuously. Once the branch changes, the dimensions of the Ritz basis and all subsequent formulas change.

The correct certificate is not a derivative of the selected singular vectors. It is an open ball on which the inequalities defining the width retain their signs. Weyl's singular-value inequality makes this radius explicit and dimension free [Stewart and guang Sun, 1990].

## 2 Sharp clipped branch radius

Let $C$ be a nonzero operator with singular values $s_1 \geq \cdots \geq s_M$. Fix a relative threshold $\tau \geq 0$, a minimum width $m$, and maximum width $M$. The policy counts ratios above threshold and clips the result into $[m, M]$.

**Theorem 1** (Threshold branch ball)**.** *Let $\widetilde{C}$ satisfy $\|\widetilde{C} - C\| \leq \varepsilon$. If $\varepsilon < \varepsilon_*$, where*

$$\varepsilon_* = min\left(\left\{\frac{s_w - \tau s_1}{1+\tau} : w > m\right\} \cup \left\{\frac{\tau s_1 - s_{w+1}}{1+\tau} : w < M\right\}\right),$$

*then the clipped threshold policy selects the same width $w$ for $\widetilde{C}$. Empty sides of the minimum are omitted.*

*Proof.* Weyl's inequality gives $|\widetilde{s}_j - s_j| \leq \varepsilon$. For a non-minimum selected width, the last retained ratio remains above threshold if

$$s_w - \varepsilon > \tau(s_1 + \varepsilon),$$

which is exactly the first radius condition. For a non-maximum width, the first omitted ratio remains below threshold if

$$s_{w+1} + \varepsilon < \tau(s_1 - \varepsilon),$$

which is the second. Together they preserve the count after clipping. When $w = m$, downward crossing of the $m$th value cannot reduce the clipped width; when $w = M$, upward motion beyond the recorded block cannot increase it. Those inequalities are therefore unnecessary. □

**Proposition 2** (Sharpness)**.** *Each finite term in $\varepsilon_*$ is best possible using only the displayed singular values and an operator-norm perturbation.*

*Proof.* For a retained wall, take a diagonal matrix and decrease $s_w$ by $\varepsilon$ while increasing $s_1$ by $\varepsilon$. At the stated radius their ratio equals $\tau$. For an omitted wall, increase $s_{w+1}$ and decrease $s_1$ by the same amount. Again equality occurs at the formula. An arbitrarily larger perturbation crosses the branch. These diagonal changes have operator norm $\varepsilon$. □

At exact contact $s_j = \tau s_1$, the open radius is zero. No continuous single-valued branch certificate exists there; a set-valued policy or a tie-breaking convention would be required.

## 3 From projectors to the cross operator

For a Hermitian memory Gram $A$ and packet projector $P$, define the ambient projected cross

$$K(A, P) = (I - P)AP.$$

Its nonzero singular values are the same as those obtained from any orthonormal packet frame, so the threshold policy is basis invariant.

| $\tau$ | strict branches | minimum relative radius | median radius | minimum / fp64 proxy |
|---|---|---|---|---|
| $10^{-8}$ | 120/120 | $4.345 \times 10^{-9}$ | $1.347 \times 10^{-2}$ | 2388.6 |
| $10^{-6}$ | 120/120 | $3.227 \times 10^{-8}$ | $1.347 \times 10^{-2}$ | 17739 |
| $10^{-4}$ | 120/120 | $3.236 \times 10^{-6}$ | $1.337 \times 10^{-2}$ | $8.89 \times 10^{5}$ |

Table 1: Finite threshold-branch margins. The fp64 column is a local backward-error proxy, not a source-model interval enclosure.

**Theorem 3** (Projected-cross perturbation)**.** *For Hermitian $A, B$ and orthogonal projectors $P, Q$,*

$$\|K(A,P) - K(B,Q)\| \le \|A - B\| + 2\|B\|\|P - Q\|.$$

*Proof.* Insert $K(B,P)$ and use

$$K(A,P) - K(B,P) = (I-P)(A-B)P,$$

while

$$K(B,P) - K(B,Q) = (Q-P)BP + (I-Q)B(P-Q).$$

Taking norms proves the formula. □

The coefficient two is the price of moving both the input and output packet legs. This bound is intentionally projector level; it avoids arbitrary basis rotations inside the selected packet.

# 4 RH-96 branch audit

The RH-96 archive contains 120 updates at each threshold $10^{-8}, 10^{-6}, 10^{-4}$. For every step we recompute $w$ and $\varepsilon_*/s_1$ from the first four projected-cross singular values. All 360 widths agree exactly with the archive, and all 360 radii are positive.

The tight primary step is the left channel at $\sigma = 0.08$, time six. Its normalized singular values are approximately

$$(1, 0.2602, 5.456 \times 10^{-4}, 5.655 \times 10^{-9}),$$

so width three is protected mainly by the omitted fourth value's distance below $10^{-8}$. The margin is positive but far smaller than a typical branch radius in the archive.

For numerical context only, we compare each relative radius with $128un$, where $u$ is binary64 unit roundoff and $n$ the ambient dimension. Every decision passes by at least three orders of magnitude. This confirms that the archived width pattern is not a floating tie. It does not certify the errors inherited from an interval source state and moving packet.

# 5 Consequence and claim boundary

The exact condition for a validated update is now modular. If a snapshot and packet enclosure gives

$$\delta_A + 2\|\widehat{A}\|\delta_P < \varepsilon_*,$$

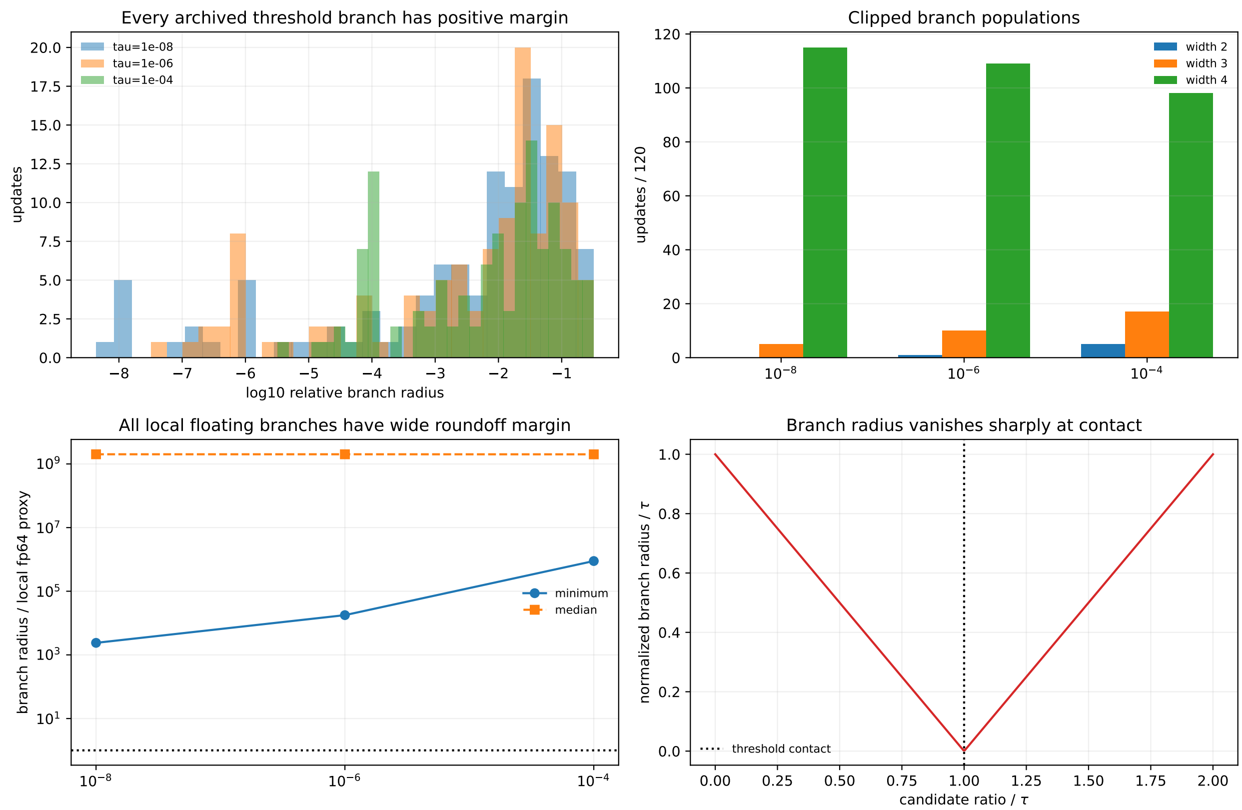


Figure 1: Branch-radius distributions, width populations, local roundoff margins, and the sharp loss of stability at threshold contact.

then Theorems 3 and 1 freeze the adaptive width; one may then enclose the fixed-dimensional Ritz/polar branch. If this inequality fails, the archived width cannot simply be assumed.

The broad coarse packet ball of RH-142 cannot yet be inserted blindly into this inequality, and no temporal source-to-cross interval radii have been computed. We have not propagated all recursive packet updates, proved a uniform threshold margin, closed controlled viability or the normalized-base liminf, established Stage A, constructed a Hilbert–Polya operator, identified zeta zeros, or proved the Riemann Hypothesis.

# References

G. W. Stewart and Ji guang Sun. *Matrix Perturbation Theory.* Academic Press, 1990.